

Problem set 1

Mathematical background / Euclidean space and motions

Modern Robotics

2013

1. Coordinate transformations

Consider an object moving with a speed $\vec{v}$, on which a force $\vec{F}$ is applied in the opposite direction (this could for instance be a friction force). $\vec{v}$ and $\vec{F}$ are depicted in Figure 1.

The velocity can be expressed as an array of numbers using the indicated coordinate frame $\bar{\Psi}$. For simplicity, the force $\vec{F}$ is also expressed as a row vector (array of numbers) $\bar{F}$, but beware that it is a co-vector and not a vector:

$$\bar{v} = 2 \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix} \quad \bar{F} = \begin{pmatrix} -\cos \alpha \\ -\sin \alpha \end{pmatrix} \quad (1)$$

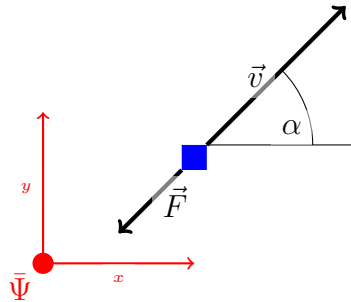


Figure 1: A moving object with a force acting on it. Note that, while both $\vec{F}$ and v are depicted using arrows, only one is a vector whilst the other is a co-vector!

Consider a second, conveniently oriented frame Ψ and calculate the matrix A that maps $\bar{v}$ into the new frame, i.e.

$$v = A\bar{v} \quad (2)$$

Then show how this transformation holds for F vs. $\bar{F}$. (Hint: you can find this transformation by using power-continuity in both frames, i.e. “ $P = \bar{P}$ ”.)

Do you notice anything peculiar?

2. Tilde form

The tilde form as introduced in the lectures is a 3×3 skew-symmetric matrix for which holds (with v and w vectors of length 3):

$$v \wedge w = \tilde{v}w \tag{3}$$

Why does this only work in a 3-dimensional world?¹

3. Skew-symmetry of $\tilde{\omega}$

According to the theory, the transport of the vector $\dot{R}$ back to the identity (by applying $R^\top$) yields $\tilde{\omega}$, a skew-symmetric matrix. Prove that $R^\top \dot{R}$ is indeed skew-symmetric.

4. Composition of H-matrices

Consider three 2D coordinate frames Ψ_1, Ψ_2 and Ψ_3 . The coordinate transformation from Ψ_1 to Ψ_2 consists of a rotation of 30° , and the coordinate transformation from Ψ_2 to Ψ_3 consists of a translation of $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$. Derive the 2D homogeneous matrices H_2^1 and H_3^2 representing these transformations, as well as H_3^1 , the composition of these two transformations.

¹Actually, in higher dimensions you can still make it work, but there the bivectors associated with the skew-symmetric matrix have redundant degrees of freedom, so they are not uniquely related to the vectors.